

UNIT TEST REPORT

By Firmgen

Report identity

Project	c5_board_validation_demo
Target	firmware (verified off target, no MCU hardware in the loop)
Report ID	UTR-03BE4CC0
Run ID	03be4cc0-61e9-40dc-86e7-4b351bebbacc
Execution date	28 September 2026, 11:05 UTC
Prepared by	Firmgen automated unit-test agent
Cases executed	51 across 20 test file(s)

1. Test Objective

This activity executed host unit tests across board integration, LEDs and user input; HTTP weather and time services; MQTT communication; UART console and diagnostics; and Ethernet configuration and network validation. It established passing evidence for 42 of 51 cases and for 2 of 9 traced requirements, while verifying 11 of the 54 enumerated firmware units. The results identify which requirement cases have executed evidence and which need further verification or assertion review.

2. Environment and Scope

Source under test	c5_board_validation_demo
Compile-time defines	UNIT_TEST, HOST_TEST, UNITY_INCLUDE_EXEC_TIME
Test parallelism	8 thread(s)
Test framework	Ceedling 1.1.6 (Unity + CMock)
Build system	Ceedling 1.1.6
Compiler	C:\Users\sujit\AppData\Local\craftifai\toolchains\mingw\16.2.0posix-14.0.0-ucrt-r1\bin\gcc.exe 16.2.0
Host environment	FirmGen-managed toolchain (ruby 3.3.12-1) on Windows-11
Execution mode	Off target - compiled for the build architecture, no MCU hardware in the loop; executed directly on the build host, not in a container
Test cases executed	51 across 20 test file(s)

3. Test Harness and Reproducibility

Tests were built and run with Ceedling 1.1.6 using Unity and CMock, with UNIT_TEST, HOST_TEST and UNITY_INCLUDE_EXEC_TIME defines and 8-thread parallelism. The measured environment was a FirmGen-managed Ruby 3.3.12-1 toolchain on Windows 11 using GCC 16.2.0, and execution was directly on the build host, off target and without MCU hardware. The environment identifies CMock as available but does not specify which individual dependencies were mocked or faked.

Reproduction step	Command	Result
Configure	`ceedling project.yml (generated per run)`	● PASS
Build and link	`ceedling test:build_only`	● PASS
Execute tests	`ceedling test:all`	● PASS

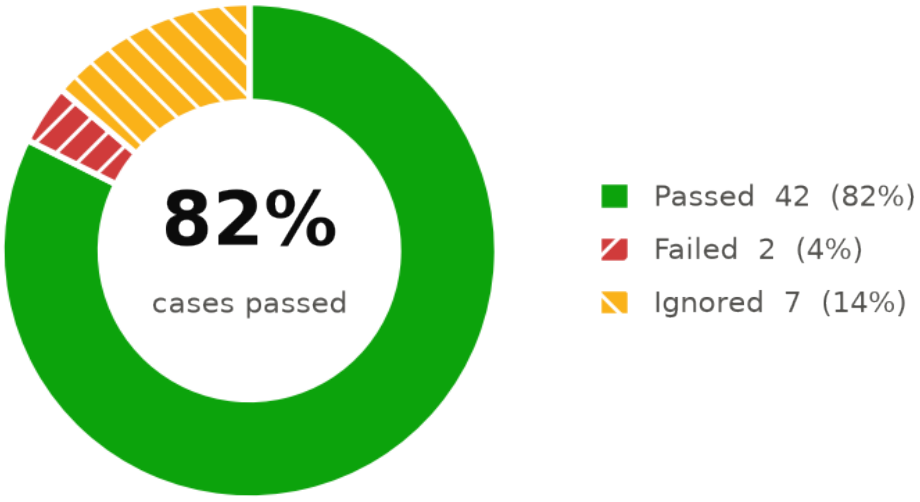
Evidence shipped with this run: executed_tests.xlsx, run.txt

4. Execution Summary

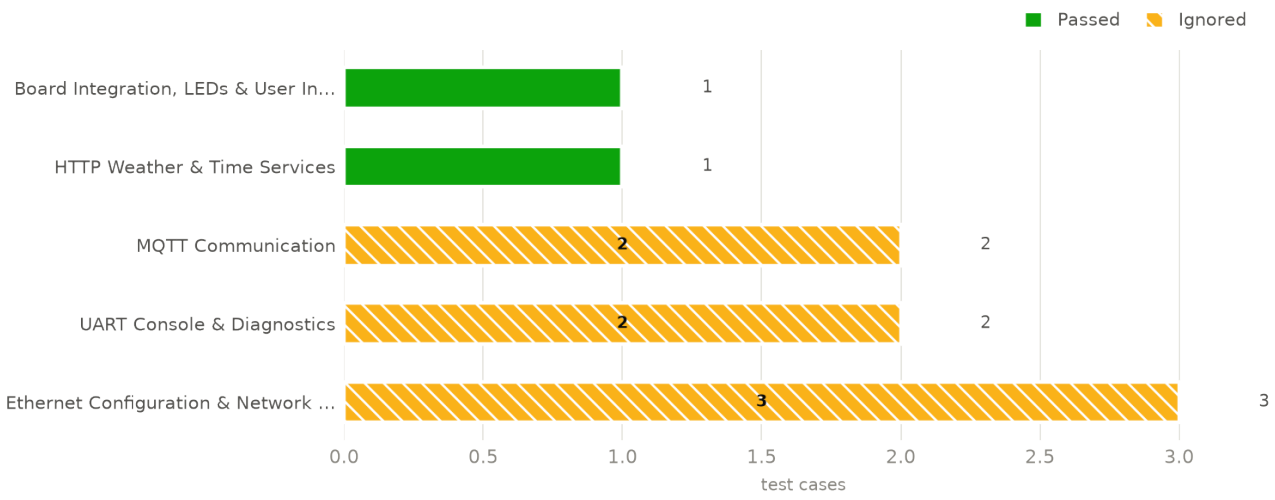
42	82%	51	2
Cases passed	Case pass rate	Cases executed	To address

Metric	Result
Cases passed	42
Case pass rate	82.4%
Requirements fully passing	2
Requirements traced to test evidence	9
Total test cases	51
Test files executed	20
Requirement-driven cases	9
Firmware-coverage cases	42
Cases awaiting a verdict	7
Cases with a failure to address	2
Requirements awaiting a verdict	7
Requirements with a failure to address	0
Total execution time	15,532 ms

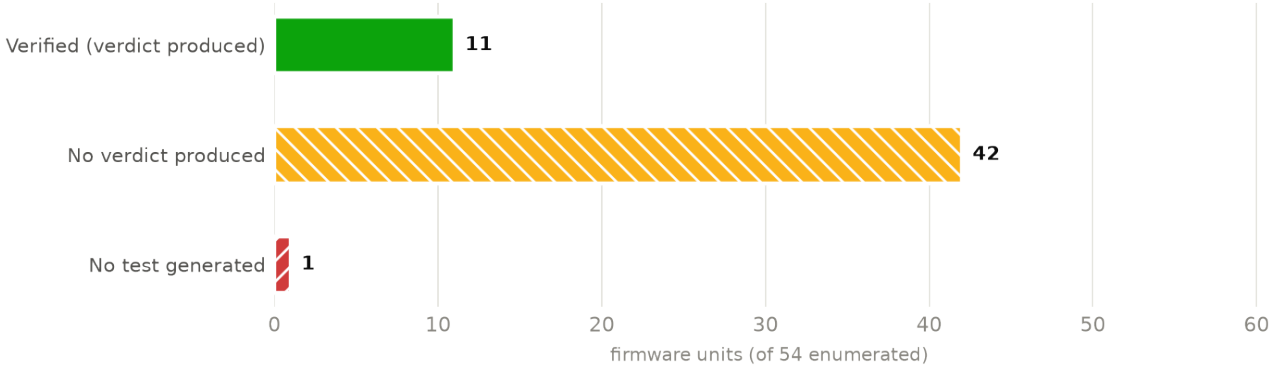
Outcome of every executed case



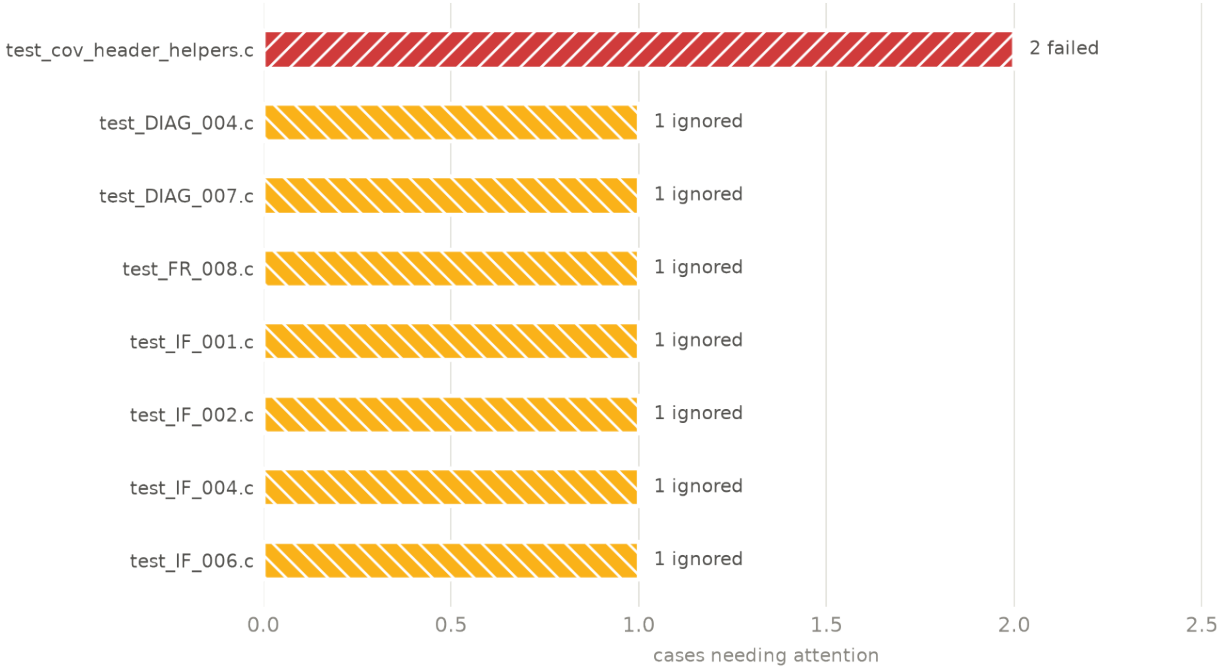
Outcomes by requirement category, strongest first



Firmware units verified by this suite



Where to focus next



5. Traceability

Executed evidence supports 2 of the 9 traced requirements; 7 requirement cases are not yet verified, and none are classified as requirements to address. The remaining requirement criteria therefore need further observable implementation paths or suitable test interfaces before they can receive a verdict.

Requirements traced to executed cases

Req. ID	Category	Module / symbol	Test file	Cases	Result
FR-006	Board Integration, LEDs & User In…	app_process_task +1 more	test_FR_006.c	1	● PASSED
FR-011	HTTP Weather & Time Services	APP_HTTP_WEATHER_HOST +2 more	test_FR_011.c	1	● PASSED
DIAG-004	Ethernet Configuration & Network…	AppLwipNetifExtCallbackFn +1 more	test_DIAG_004.c	1	● IGNORED
DIAG-007	UART Console & Diagnostics	app_process_task +1 more	test_DIAG_007.c	1	● IGNORED
FR-008	MQTT Communication	mqtt_platform_publish_telemetry	test_FR_008.c	1	● IGNORED
IF-001	UART Console & Diagnostics	AppLoggerHeader +2 more	test_IF_001.c	1	● IGNORED
IF-002	Ethernet Configuration & Network…	AppLwipNetifExtCallbackFn +3 more	test_IF_002.c	1	● IGNORED
IF-004	MQTT Communication	mqtt_platform_publish_telemetry	test_IF_004.c	1	● IGNORED
IF-006	Ethernet Configuration & Network…	AppLwipNetifExtCallbackFn +1 more	test_IF_006.c	1	● IGNORED

Firmware units traced to executed cases

Firmware unit	Kind	Test file	Cases	Result
board	header	test_cov_header_board.c	2	● PASSED
board_name	function	test_cov_function_board_name.c	1	● PASSED
button_logic_get_shared_state	function	test_cov_function_button_logic_get_shared_state.c	1	● PASSED
button_logic_init	function	test_cov_function_button_logic_init.c	3	● PASSED
button_logic_set_shared_state	function	test_cov_function_button_logic_set_shared_state.c	2	● PASSED
button_logic_toggle_state	function	test_cov_function_button_logic_toggle_state.c	3	● PASSED
button_logic_update	function	test_cov_function_button_logic_update.c	10	● PASSED
logger	header	test_cov_header_logger.c	4	● PASSED
network_status_is_ready	function	test_cov_function_network_status_is_ready.c	2	● PASSED
network_status_set_ready	function	test_cov_function_network_status_set_ready.c	2	● PASSED
helpers	header	test_cov_header_helpers.c	12	● FAILED

6. Failure, Ignored Case, and Root-Cause Record

Scope / cases	Observed result	Suspected cause	Severity	Disposition
UART output observation and configuration DIAG-004, DIAG-007, IF-001	These cases were ignored because the supplied code does not provide an observable UART capture or writer interface for the required output checks; the serial-configuration case also lacks UART initialization, baud/framing configuration, and serial-reception observation.	The supplied implementation and test interfaces do not expose the UART output and configuration needed to evaluate these criteria.	● Low	Open; provide an observable UART writer or capture interface and the relevant initi alization/configu ration path, then add or enable tests for the specified output and serial setti…
Telemetry publication scheduling FR-008	The case was ignored because the supplied source shows no caller or scheduler for mqtt_platfo rm_publish_telemetry from which publication intervals could be observed.	The publication timing behavior is outside the visible implementation path supplied for this test.	● Low	Open; expose or supply the scheduling path and a controllable time source so adjacent publication intervals can be tested.
MQTT publication topic setup IF-004	The case was ignored because the publication branch depends on private client and connection state that cannot be set through a public setup or connection path shown in the supplied source.	The required MQTT state is not accessible through the available test setup path.	● Low	Open; provide a testable public setup path or suitable injection seam for client and connection state, then verify the publication topic.
Network validation source availability IF-002, IF-006	Both cases were ignored: the cited lwIP and application sources needed to compare the DHCP ACK with the assigned address were absent, and the cited NTP and network-validation functions needed to observe UDP port 123 and separate results were not present.	The supplied archive does not include the implementation paths required to observe these network-validation criteria.	● Low	Open; include or expose the relevant DHCP, NTP and validation implementation paths, then add tests for address assignment, NTP exchange and separate validation results.
Header helper assertion expectations no ids cited	Both cases failed: test_helpers_05 expected 1 and observed 2; test_helpers_10 expected 5 and observed 6.	The expected values do not match the observed helper results; the supplied evidence does not establish whether the test expectations or helper behavior need correction.	● Medium	Open; compare each expected value with the intended helper behavior, then update the test or implementation as appropriate and rerun both cases.

7. Limitations and Exclusions

This report states what the executed cases established. The boundaries below mark where off-target unit testing ends and other verification activities begin - they are the scope of this activity, not defects in it.

- These off-target host tests do not establish electrical behavior; target-level or hardware-in-the-loop verification is needed for that boundary.
- They do not establish interrupt latency or RTOS scheduling behavior; those require verification in an environment that exercises the target and its runtime.
- They do not establish UART reception by a physical serial terminal or validate physical baud and framing behavior.
- They do not verify the network exchanges or timing criteria for which the supplied implementation or observable test interfaces were unavailable.

8. Verification Status and Conclusion

Decision field	Conclusion
Verified scope	42 of 51 executed case(s) passed, covering 2 requirement(s) in full across 20 test file(s).
Outstanding items	Review the two helper assertion expectations against intended behavior and rerun those cases. Add observable UART and MQTT setup or scheduling paths, and provide the DHCP and NTP/network-validation implementation paths needed to test the seven cases that remain without a verdict.
Conclusion	The activity provides executed passing evidence across the tested firmware units, including 2 traced requirements, while distinguishing those results from cases that could not be evaluated. The two failed assertions and seven ignored cases have specific next steps: reconcile helper expectations and supply the missing implementation visibility or test interfaces. The results support continued, targeted verification without making claims beyond the host-test evidence.

Generated automatically from the executed test results of run 03be4cc0-61e9-40dc-86e7-4b351bebbacc. The figures in this document are computed from executed_tests.xlsx; no count in it was written by a language model.

9. Appendix: Complete Case Record

Every case this run executed appears below, one row each - 51 in total, R1-R9 from the requirement suite and C1-C42 from the firmware-coverage suite. These are the rows every figure in Section 4 was counted from.

Case	Identifier	Test function	Result
R1	DIAG-004	test_DIAG_004_01	IGNORED
R2	DIAG-007	test_DIAG_007_01	IGNORED
R3	FR-006	test_FR_006_01	PASSED
R4	FR-008	test_FR_008_01	IGNORED
R5	FR-011	test_FR_011_01	PASSED
R6	IF-001	test_IF_001_01	IGNORED
R7	IF-002	test_IF_002_01	IGNORED
R8	IF-004	test_IF_004_01	IGNORED
R9	IF-006	test_IF_006_01	IGNORED
C1	board	test_board_01	PASSED

Case	Identifier	Test function	Result
C2	board	test_board_02	PASSED
C3	board_name	test_board_name_01	PASSED
C4	button_logic_get_shared_state	test_button_logic_get_shared_state_01	PASSED
C5	button_logic_init	test_button_logic_init_01	PASSED
C6	button_logic_init	test_button_logic_init_02	PASSED
C7	button_logic_init	test_button_logic_init_03	PASSED
C8	button_logic_set_shared_state	test_button_logic_set_shared_state_01	PASSED
C9	button_logic_set_shared_state	test_button_logic_set_shared_state_02	PASSED
C10	button_logic_toggle_state	test_button_logic_toggle_state_01	PASSED
C11	button_logic_toggle_state	test_button_logic_toggle_state_02	PASSED
C12	button_logic_toggle_state	test_button_logic_toggle_state_03	PASSED
C13	button_logic_update	test_button_logic_update_01	PASSED
C14	button_logic_update	test_button_logic_update_02	PASSED
C15	button_logic_update	test_button_logic_update_03	PASSED
C16	button_logic_update	test_button_logic_update_04	PASSED
C17	button_logic_update	test_button_logic_update_05	PASSED
C18	button_logic_update	test_button_logic_update_06	PASSED
C19	button_logic_update	test_button_logic_update_07	PASSED
C20	button_logic_update	test_button_logic_update_08	PASSED
C21	button_logic_update	test_button_logic_update_09	PASSED
C22	button_logic_update	test_button_logic_update_10	PASSED
C23	helpers	test_helpers_01	PASSED
C24	helpers	test_helpers_02	PASSED
C25	helpers	test_helpers_03	PASSED
C26	helpers	test_helpers_04	PASSED
C27	helpers	test_helpers_05	FAILED
C28	helpers	test_helpers_06	PASSED
C29	helpers	test_helpers_07	PASSED
C30	helpers	test_helpers_08	PASSED
C31	helpers	test_helpers_09	PASSED
C32	helpers	test_helpers_10	FAILED
C33	helpers	test_helpers_11	PASSED
C34	helpers	test_helpers_12	PASSED
C35	logger	test_logger_01	PASSED
C36	logger	test_logger_02	PASSED
C37	logger	test_logger_03	PASSED
C38	logger	test_logger_04	PASSED
C39	network_status_is_ready	test_network_status_is_ready_01	PASSED
C40	network_status_is_ready	test_network_status_is_ready_02	PASSED

Case	Identifier	Test function	Result
C41	network_status_set_ready	test_network_status_set_ready_01	● PASSED
C42	network_status_set_ready	test_network_status_set_ready_02	● PASSED